

William Muckelroy III

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EDUCATION

University of Pittsburgh

Bachelor's of Science in Engineering, Computer Engineering

Major GPA: 3.689

Aug. 2020 – Dec. 2025

Pittsburgh, PA

Dean's Honor List (All Semesters)

Relevant Coursework:

Data Structures & Algorithms; Computational Modeling & Simulations; Robotics Studio; Image Processing; Robotics Controls; Optimization Theory; F1Tenth Robotics (self-study); Signals, Systems & Probability; Embedded Processors & Interfacing

PUBLICATIONS & PRE-PRINTS

- W. Muckelroy III, M. Alsakabi, J. M. Dolan, O. K. Tonguz, "The Impact of 2D Segmentation Backbones on Point Cloud Predictions Using 4D Radar." Working Paper. [preprint here \(https://arxiv.org/abs/2509.19644\)](https://arxiv.org/abs/2509.19644).
- A. Saba, et al., "Fast and Modular Autonomy Software for Autonomous Racing Vehicles." IEEE Transactions on Field Robotics, vol. 4, pp. 1-45, Jan. 2024, doi: [10.55417/fr.2024001](https://doi.org/10.55417/fr.2024001).

ACADEMIC RESEARCH EXPERIENCE

Stanford AI Lab (SAIL) | Interactive Perception & Robot Learning (IPRL) Lab

Post-Bacc Visiting Researcher | Advised by [Prof. Jeannette Bohg](#) & [Dr. Cherie Ho](#)

March 2026 – Present

Stanford, CA

- Developing a persistent, object-centric memory system linking RGB-D perception with adaptive task planning for mobile manipulators. This system uses a real-time scene-graph that synchronizes perception, action, and memory, updating as the robot explores and interacts with its environment. We also leverage VLMs grounded with physical awareness to predict object affordances and choose appropriate motion primitives for interaction.

Carnegie Mellon University (CMU) - Robotics Institute Summer Scholars (RISS)

Research Experience for Undergraduates (REU) | [DRIVE Lab](#) | Advised by [Dr. John Dolan](#)

June 2025 – Present

Pittsburgh, PA

- Decomposed a Neural Network used for predicting 3D point clouds from raw 4D Radar signals into a 2D Segmentation Backbone and Temporal Coherence Network that I modified to yield a 24% improvement in bidirectional Chamfer distance.
- Exploring the use of LiDAR point cloud reconstruction techniques and incorporating spatiotemporal information for improved resolution and accuracy in point cloud detections using 4D Radar.
- Exploring multi-agent collaborative perception and game theory for high-level cognition in V2X/V2V autonomous systems.

MIT-Pitt-RW Indy Autonomous Team

Perception Engineer | [Indy Autonomous Challenge](#) | Advised by [Dr. Sam Dickerson](#)

Oct. 2022 – Aug. 2024

Pittsburgh, PA

- Completed field testing & competitions at tracks across the country where we tested our algorithms and autonomy stack, of which, I worked on developing the perception stack by designing new ML pipelines to clean, process, label, and train a YOLOv5 vehicle detection model capable of detecting cars at speeds greater than 150 mph and from a distance beyond 100 meters and investigating new methods for camera + LiDAR sensor fusion with extended Kalman filters (EKFs).

INDUSTRY RESEARCH & ENGINEERING EXPERIENCE

Paint the Canvas: Low-Cost Hardware for 3D Scene Reconstruction using 3D Gaussian Splatting

Senior Design Capstone Project | Project & Research Lead | *3rd Place Overall*

Aug. 2025 - Dec. 2025

Pittsburgh, PA

- Led a team of 4 to design a sub-\$200 handheld sensor suite for autonomous 3D scene reconstruction (custom enclosure + PCB), built a ROS2-based GUI linking the handheld and laptop basestation, and integrated 3D Gaussian Splatting, NeRF, and Sparse Voxel Rasterization to compare scene training/rendering performance.

Google Labs

Software Engineering Intern | Advised by [Dr. Zhu](#), [Dr. Bloniarz](#), and [Ben Murdoch](#)

May 2024 – Aug. 2024

Boulder, CO

- Added image modality to a newly developed chain-of-thought multi-hop large language model (LLM) agent used for deep research tasks, where I was able to improve model evaluation metrics by around 100% against an internal SOTA model while developing groundedness evaluation pipelines to ensure no hallucinations and semantic understanding of tasks & prompts.

Amazon Lab126

Software Engineering Intern | [Project Astro](#) | Advised by [Dr. Apaar Sadhwani](#)

Sept. 2023 – Dec. 2023

Sunnyvale, CA

- Built a custom reward schema and path planning algorithm, allowing Project Astro to increase its floor plan coverage from 40% to over 90% by discretizing the floor plan into areas of high and low rewards based on Astro's current functionality (finding a person vs. avoiding clutter), allowing Astro to interact with people and their home environment more naturally.

Google Labs

Software Engineering Intern | [Project Starline \(Beam\)](#) | Advised by [Dr. Steve Cholewiak](#)

June 2023 – Sept. 2023

Mountain View, CA

- Designed a testing infrastructure using an UR5e robot arm, an artificial 3D-printed head, and cameras for eyes that was meant to quantify 3D visual crosstalk in a lenticular display. This system was robust enough to test and calibrate the head-tracking accuracy, motion-to-photon latency, and the Starline (now Google Beam) displays.
- Research scientists guided me to learn more about the display and camera technologies, where I performed literature reviews and conducted various experiments on the tuning of optics and ensured reliable and consistent testing.

Tesla

Autopilot Hardware Engineer

May 2022 – May 2023

Austin, TX

- Designed software that used JSON/yaml representations of memory controller configurations and memory traffic scenarios as input to then simulate the performance (access latency, bandwidth/throughput, deadlocks, etc.) of hundreds of memory controller designs to aid the Autopilot HW team to make informed design decisions with performance in mind.
- Implemented various common NN layers (convolutions, activation functions, pooling, etc.) with PyTorch to serve as ground-truth comparison against ML optimized compilers.
- Modeled the performance of data transfer and neural network algorithm inference (i.e., deformable convolutions) on Tesla's DOJO computing platform using Assembly and Python.

LEADERSHIP EXPERIENCE

National Society of Black Engineers

University of Pittsburgh Chapter's Academic Excellence Chair

May 2021 – May 2022

- Planned & executed multiple events aimed at maintaining and improving the overall GPA of the chapter. For example, organizing study events with over 200 participants across all Engineering departments where faculty and peers aided in tutoring for engineering classes ranging from the freshman year curriculum to more advanced senior curriculum.

Robotics & STEM Community Outreach

Sept. 2021 - Present

- Built pathways to STEM for kids through outreach events with my participation in the National Society of Black Engineers (NSBE), Society of Hispanic Professionals in Engineering (SHPE), Pitt, and CMU communities. This included building STEM kits, co-leading introduction to robotics events, and exposure to engineering seminars for K-12 students in Pittsburgh.
- Collaborated with Carnegie Mellon University (CMU) to help address food scarcity within the Hazelwood community in Pittsburgh, PA by aggregating and distributing resources for food acquisition.

Undergraduate Research Advocacy

In Partnership with Carnegie Mellon University Under Rachel Burcin

Sep. 2025 – Mar. 2026

- Helped develop presentations and documentation that were presented to PA Congress representatives and the head of CMU in order to advocate for continued funding and support for undergraduate research, which has proven to be a success criterion for continuation into advanced graduate degrees and innovation.

SEA (Summer Engineering Academy) Mentor

July 2021 - Aug. 2021

Undergraduate Education Researcher

- Organized events and gave talks to prepare incoming Engineering freshmen in a 2-week crash course into Pitt's 1st year Engineering curriculum
- Collaborated with multiple faculty to aid in teaching physics and calculus classes, grading assignments, and hosting office hours to assist scholars in this intensive program.
- Hosted multiple motivational talks and presentations to provide unique and uplifting perspectives to incoming scholars.

F1Tenth Robotics Course Development

Jan. 2024 – May 2024

Undergraduate Education Researcher

- Assisted in the development of a robotics F1Tenth course for the ECE Department at the University of Pittsburgh through research and participation in self-paced F1Tenth modules.

PRESENTATIONS

- **W. Muckelroy III**, K. Hu, M. Alsakabi, J. M. Dolan, O. K. Tonguz, "LiDAR to Radar: Defining the Future for Intelligent Transportation Systems." Presenting [research work](#) on behalf of DRIVE Lab at [Safety21's Technology Showcase](#), Nov. 2025.
- **W. Muckelroy III**, M. Alsakabi, J. M. Dolan, O. K. Tonguz, "Investigating the Effects of Higher Capacity Segmentation Backbones on 4D Radar Point Cloud Generation" Presented work within CMU's RISS research showcase poster session ([see here](#)) and submitted a paper to the 2025 RISS Working Papers Journal, Aug. 2025. **Selected as the RISS Finale speaker.**
- **W. Muckelroy III**, S. Dickerson, "An Ever-Evolving World With Cloud Computing And Cyber Physical Systems." Presented to faculty, staff, and students from the Swanson School of Engineering, July 2021. (1st Year REU). [Read Here](#).

- **W. Muckelroy III**, N. Maluf, R. Lehrman, “Examining The Viability Of Lithium-Ion Technology For Use In Electric Vehicles.” Submitted a paper and presented at the Swanson’s First Year Engineering Conference (FYEC), April 2021. [Read Here](#).

SKILLS

Programming Languages: Python, C + + , Java, MATLAB, C, Assembly, LaTeX
Operating Systems: Linux, MacOS, ROS(2)
Additional Skills: PyTorch, TensorFlow, Version Control (i.e., Git, Jira, Confluence, etc.)

MEMBERSHIPS

Society of Hispanic Professionals in Engineering (SHPE)	Fall 2020 - Present
National Society of Black Engineers (NSBE)	Fall 2020 - Present
Robotics & Automation Society (RAS)	Fall 2021 - Present
Pitt Computer Science Club (CSC)	Fall 2020 - Present
Institute of Electrical and Electronics Engineers (IEEE)	Fall 2021 - Present

AWARDS & HONORS

Google Undergraduate Research Scholarship	Fall 2025
Stanford SERGE Fellow	Fall 2025
RISS Undergraduate Research Fellowship Award	Summer 2025
Safety21 Research Scholarship	Summer 2025
Texaco Foundation Scholarship (\$8k/semester)	Fall 2024
Honeywell Corporate Scholarship (\$5k/year)	Fall 2024
National Society of Collegiate Scholars Geico Award (\$3k)	Fall 2021
University of Pittsburgh’s Summer Research Internship Fellowship	Summer 2021
Lockheed Martin Corporate Scholarship (\$5k/year)	Spring 2021
Raytheon Technologies Scholarship (\$5k/year)	Spring 2021
Chevron Corporate Scholarship (\$5k/year)	Spring 2021

ADDITIONAL EXPERIENCE

University of Pittsburgh	May 2021 – July 2021
REU Freshman Summer Research Internship ECE Department	Pittsburgh, PA
• Conducted research on the future impact of cloud computing and IoT on the automotive and healthcare industries. • Wrote a paper and presented my work to a panel of students and faculty at the University of Pittsburgh. • Programmed a Texas Instruments Robotics Systems Learning Kit (TI-RSLK) robot with C to follow physical tracks autonomously using IR & bumper sensors.	
University of Pittsburgh - First-Year Engineering Conference	Spring 2021
Swanson School of Engineering Freshman Engineering Research	Pittsburgh, PA
• Led a team to investigate the viability and sustainability of lithium-ion battery technology in autonomous vehicles by analyzing factors such as overall carbon footprint, life cycle analysis, and efficiency compared to internal combustion engine (ICE) vehicles.	